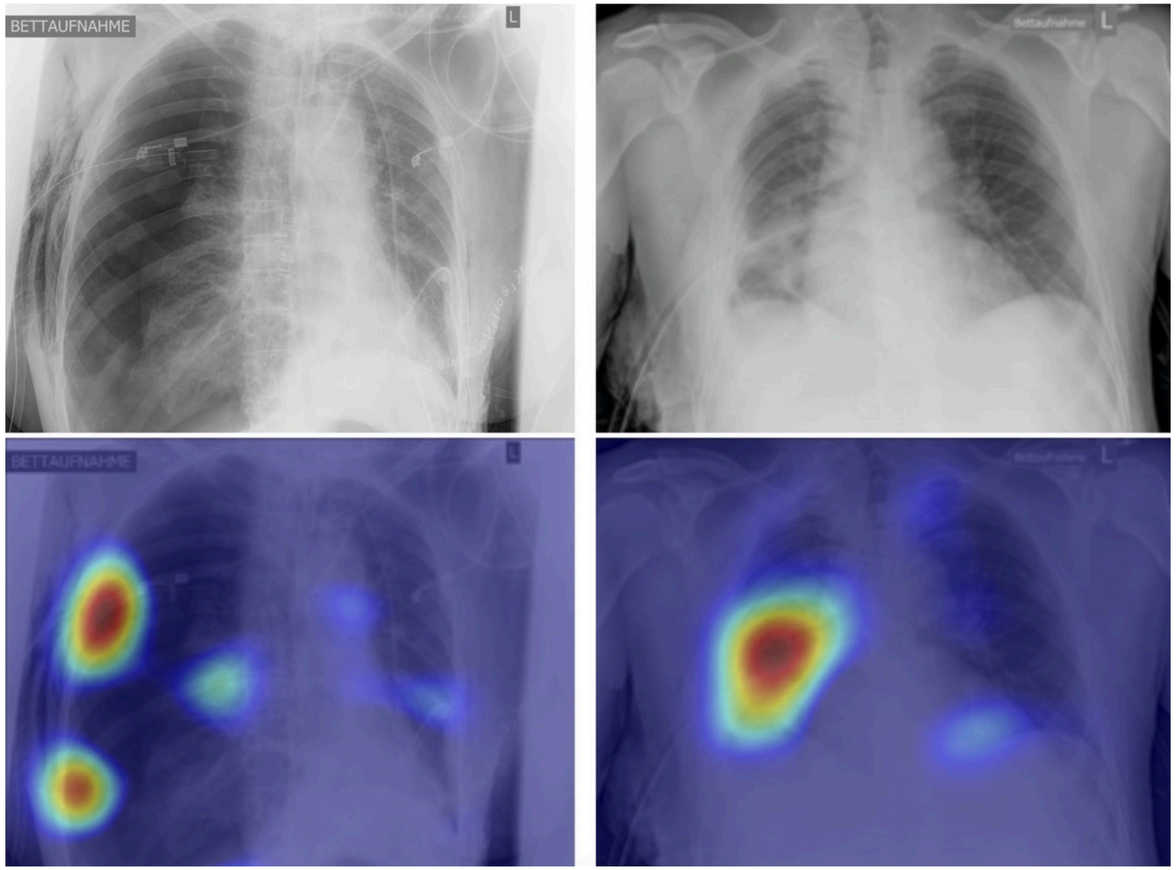
Trustworthy ML - Part 2

Generalisation Studies

Context Recap



- (More on models' tendency to learn such rules later on in the semester)

Is there a more principled approach?

Generalisation Theory

Statistical Learning Theory

The purely theoretical approach

Statistical Learning Theory

The purely theoretical approach

- **Objective:** Bound generalisation

Statistical Learning Theory

The purely theoretical approach

- **Objective:** Bound generalisation
- Probably Approximately Correct (PAC) bounds

$$\mathbb{P}\left(R(\phi) \leq \delta\right) \geq 1 - \epsilon$$

Statistical Learning Theory

The purely theoretical approach

- **Objective:** Bound generalisation
- Probably Approximately Correct (PAC) bounds

$$\mathbb{P}\left(R(\phi) \leq \delta\right) \geq 1 - \epsilon$$

generalisation performance (Risk)

Statistical Learning Theory

The purely theoretical approach

- **Objective:** Bound generalisation
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- Central to computing this is some notion of hypothesis class expressivity (complexity/capacity)

Expressivity

- **Intuition:** Occam's razor

  "It is futile to do with more what can be done with fewer."

- e.g. VC dimension, Rademacher complexity

VC Dimension

- **Intuition:** Maximum number of samples that can be classified in all possible combinations

VC Dimension

- **Class:** Axis-aligned rectangles

VC Dimension

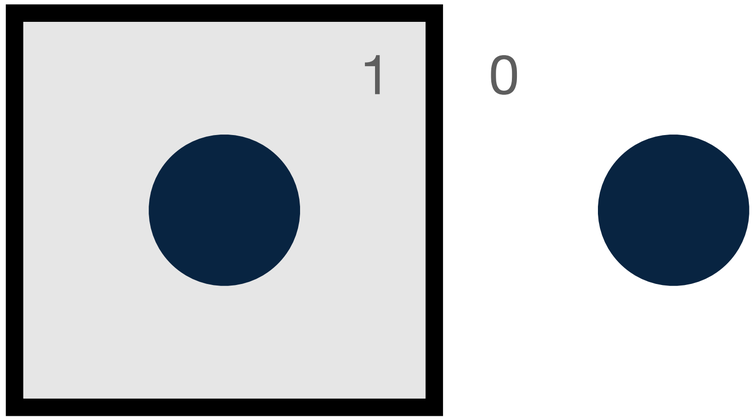
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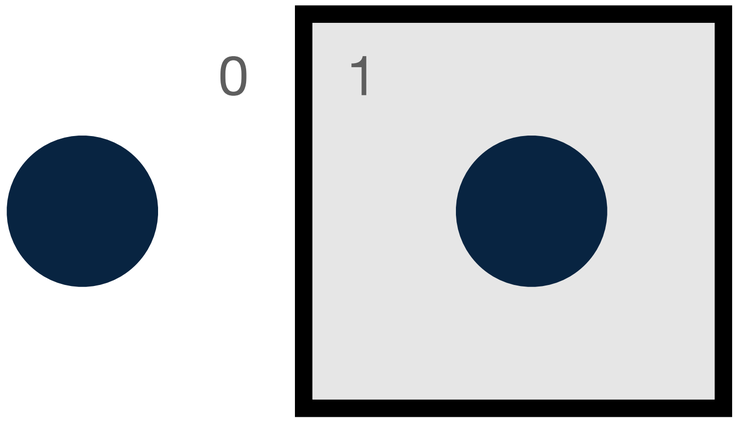
VC Dimension



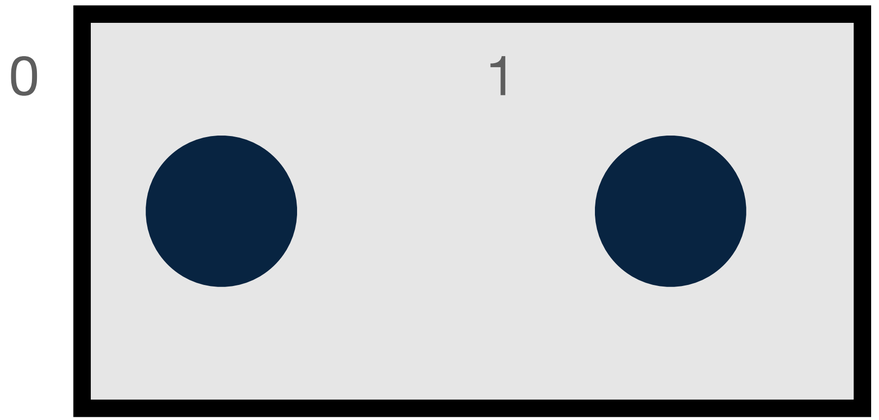
VC Dimension



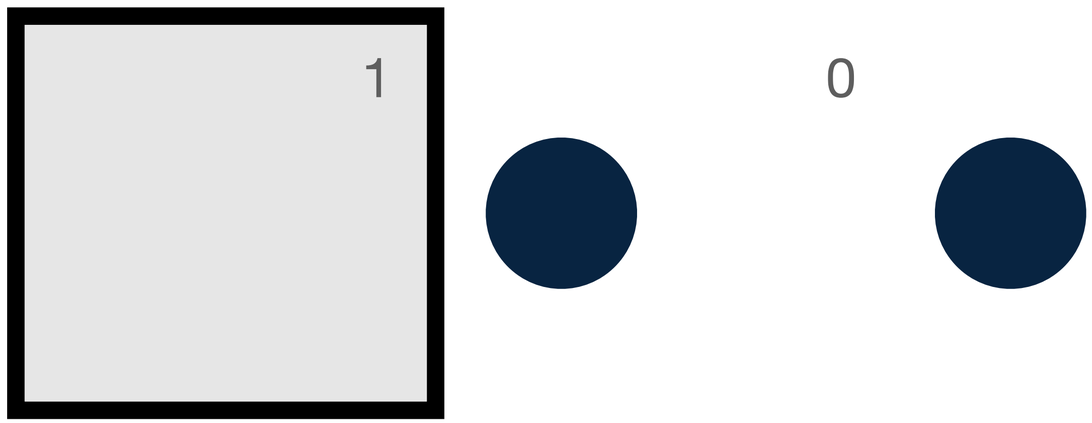
VC Dimension



VC Dimension



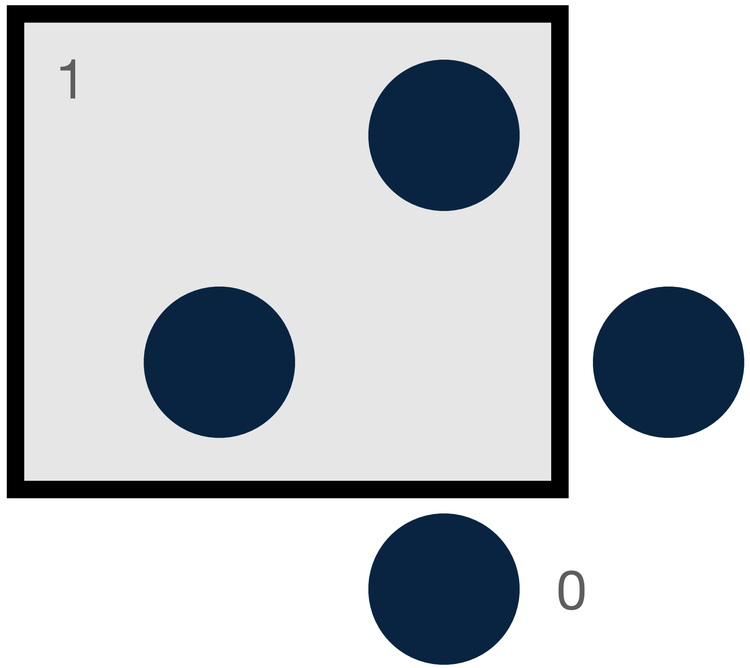
VC Dimension



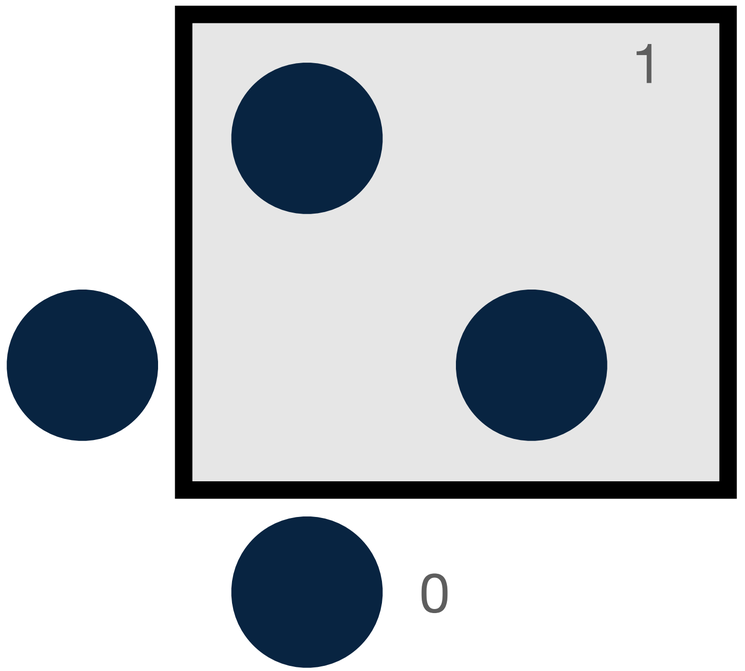
VC Dimension

VC dimension of axis-aligned rectangles is at least 2

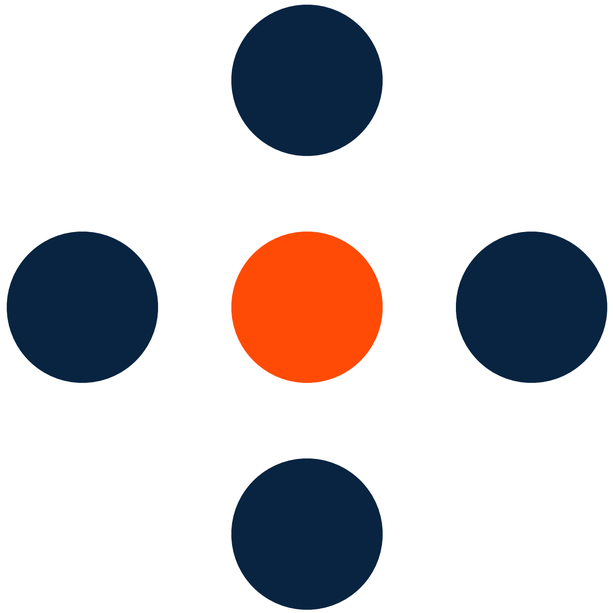
VC Dimension



VC Dimension



VC Dimension



VC Dimension

- **Class:** Axis-aligned rectangles
- **VC Dimension:** 4

Putting Things Together



Statistical Learning Theory

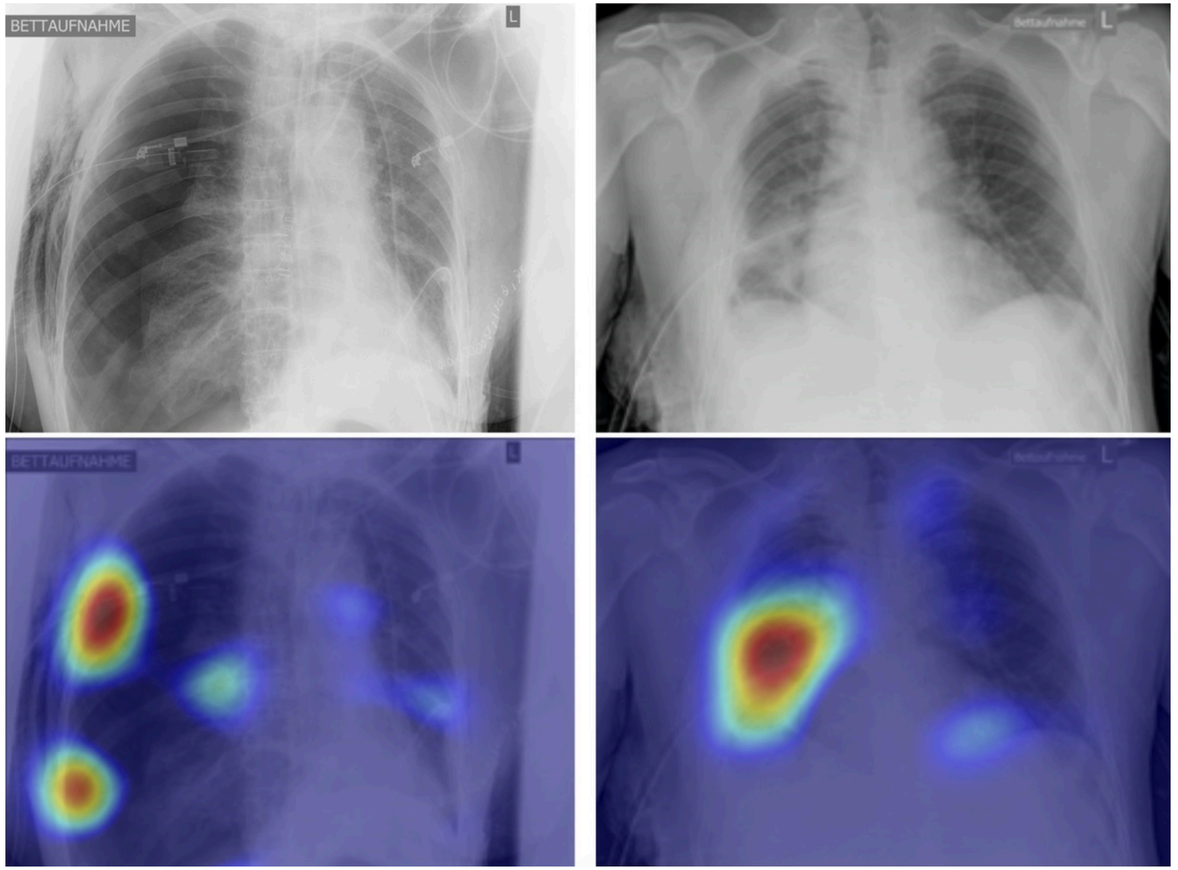
The purely theoretical approach

- Guarantees
- Rigorous

- Assumption: (i.i.d) Training and test data are drawn from the same distribution

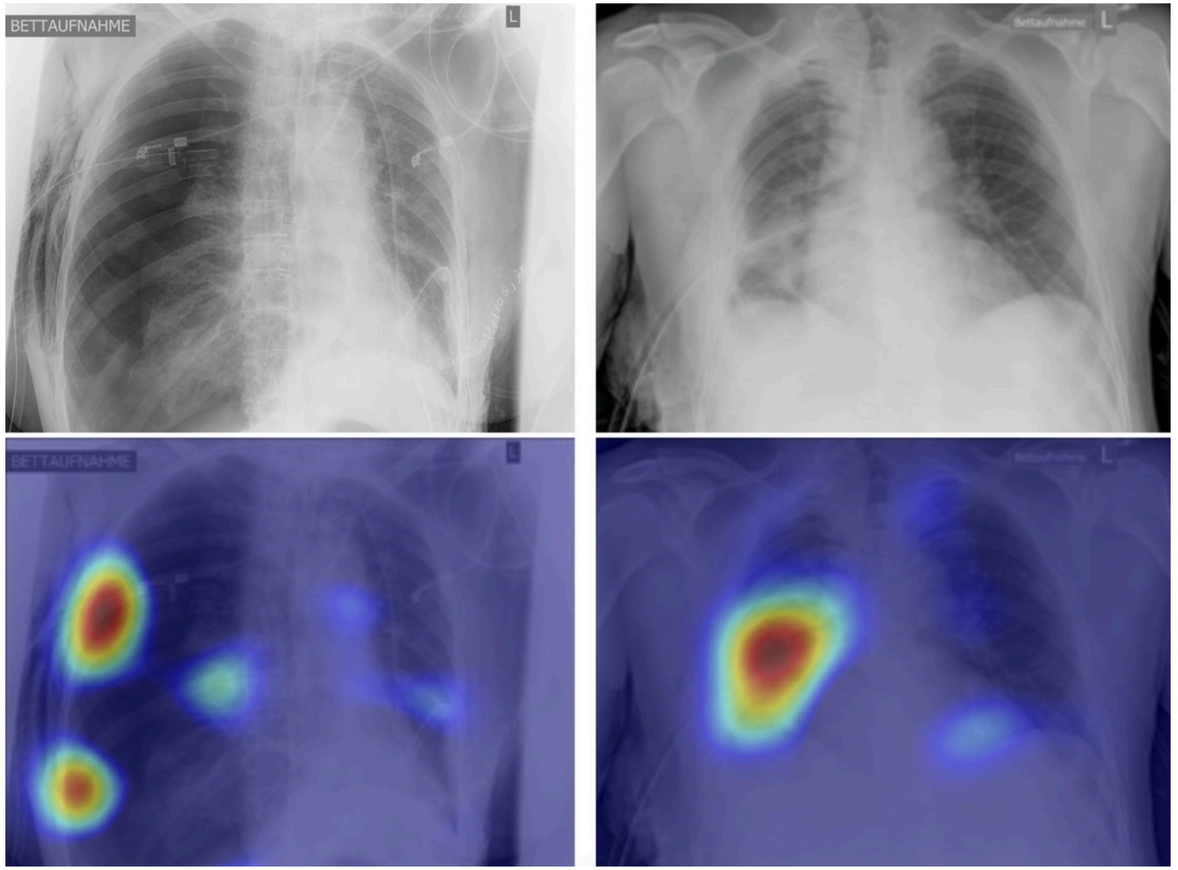
Statistical Learning Theory

The purely theoretical approach



Statistical Learning Theory

The purely theoretical approach



i.i.d???

Statistical Learning Theory

The purely theoretical approach

- Guarantees
- Rigorous

- **STRONG** Assumption: (i.i.d) Training and test data are drawn from the same distribution

Statistical Learning Theory

The purely theoretical approach

- Guarantees
- Rigorous

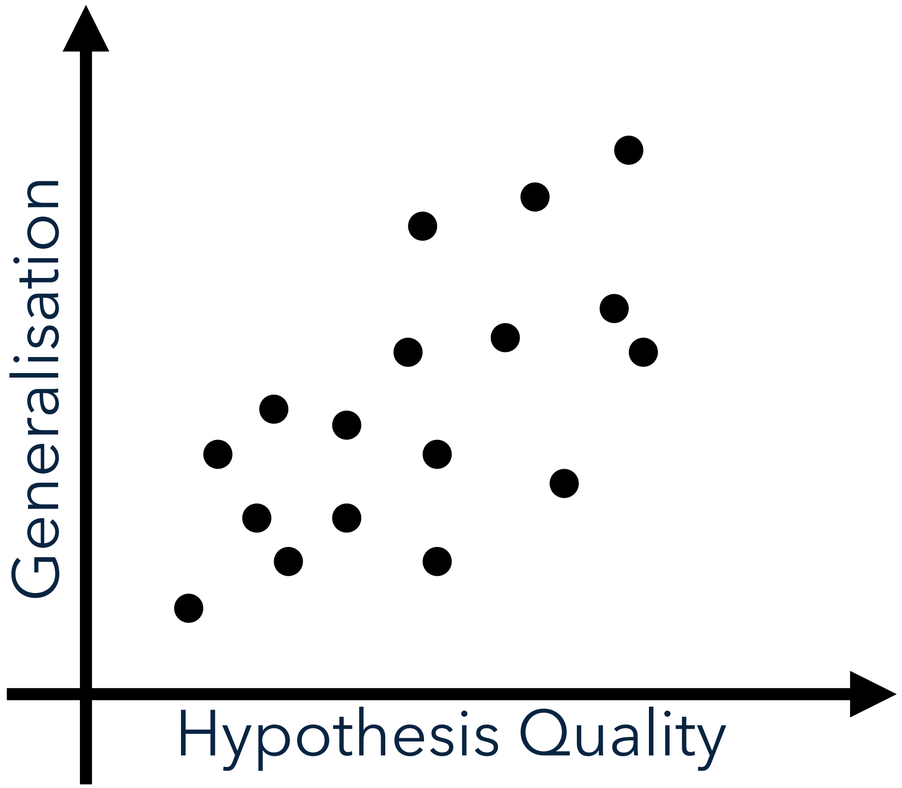
- **STRONG** Assumption: (i.i.d) Training and test data are drawn from the same distribution
- Uninformative in practice

If you want to read more on the topic, "Understanding Machine Learning: From Theory to Algorithms" by Shai Ben-David and Shai Shalev-Shwartz (video series available as well)

Empirical Theory of Generalisation

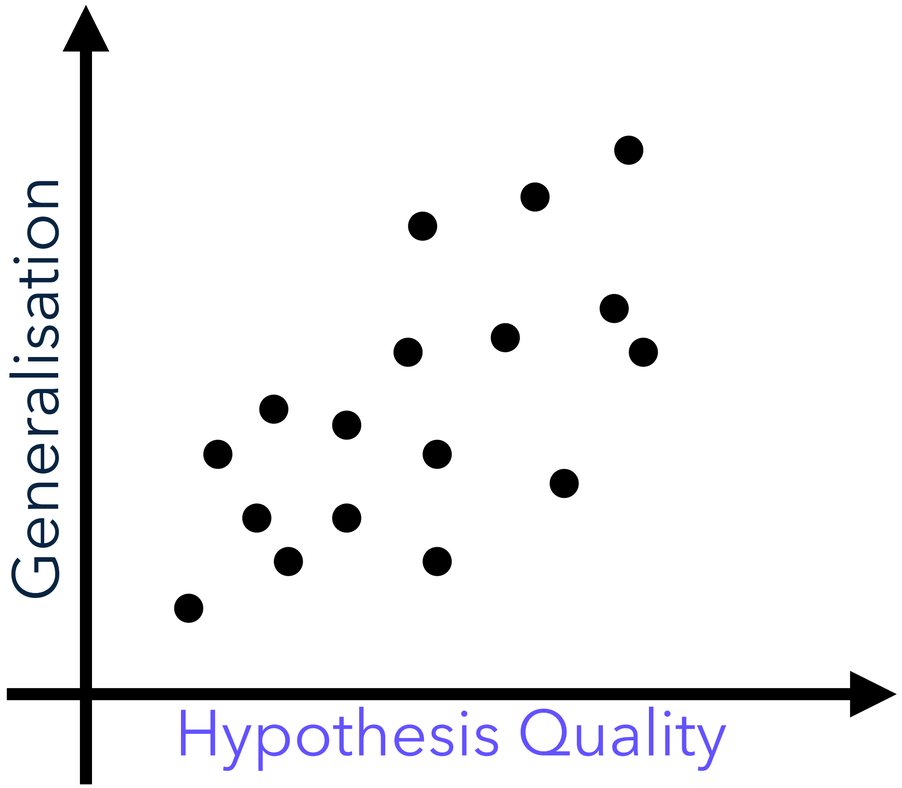
Empirical Approach

- **Objective:** Predict generalisation



Empirical Approach

- **Objective:** Predict generalisation



Expressivity?

VC dimension?

Expressivity?

VC dimension? Not good for DL

Expressivity?

Number of parameters?

Expressivity?

Number of parameters?

- Does expressivity necessarily increases with the number of parameters?

Expressivity?

Number of parameters?

- CNN expressivity is not necessarily linear in the number of parameters
- Not suitable for comparing CNN against CNN



Expressivity of CNNs

- How do we know CNNs are highly expressive? Give example of one experiment that might demonstrate this. Hint: you can talk about a well-known experiment – but you can also propose your own :)

Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning requires rethinking generalization. In International Conference on Learning Representations, 2017

Expressivity of CNNs

- ImageNet:
  - 1000 classes
  - 224 x 224 pixels
  - More than 1M images
- CNNS **can** be very expressive, but models don't have to make use of all the expressivity

Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning requires rethinking generalization. In International Conference on Learning Representations, 2017

Expressivity of a Hypothesis

Expressivity of a Hypothesis

- DNN equivalent for a hypothesis?

  Fixed set of parameters (ish, because of permutation symmetries)

  = “learned model”

- What is the expressivity of a learned model?
  - The number of parameters is not a great notion
  - How about the number of non-zero parameters? Or their magnitude?

Expressivity of a Hypothesis

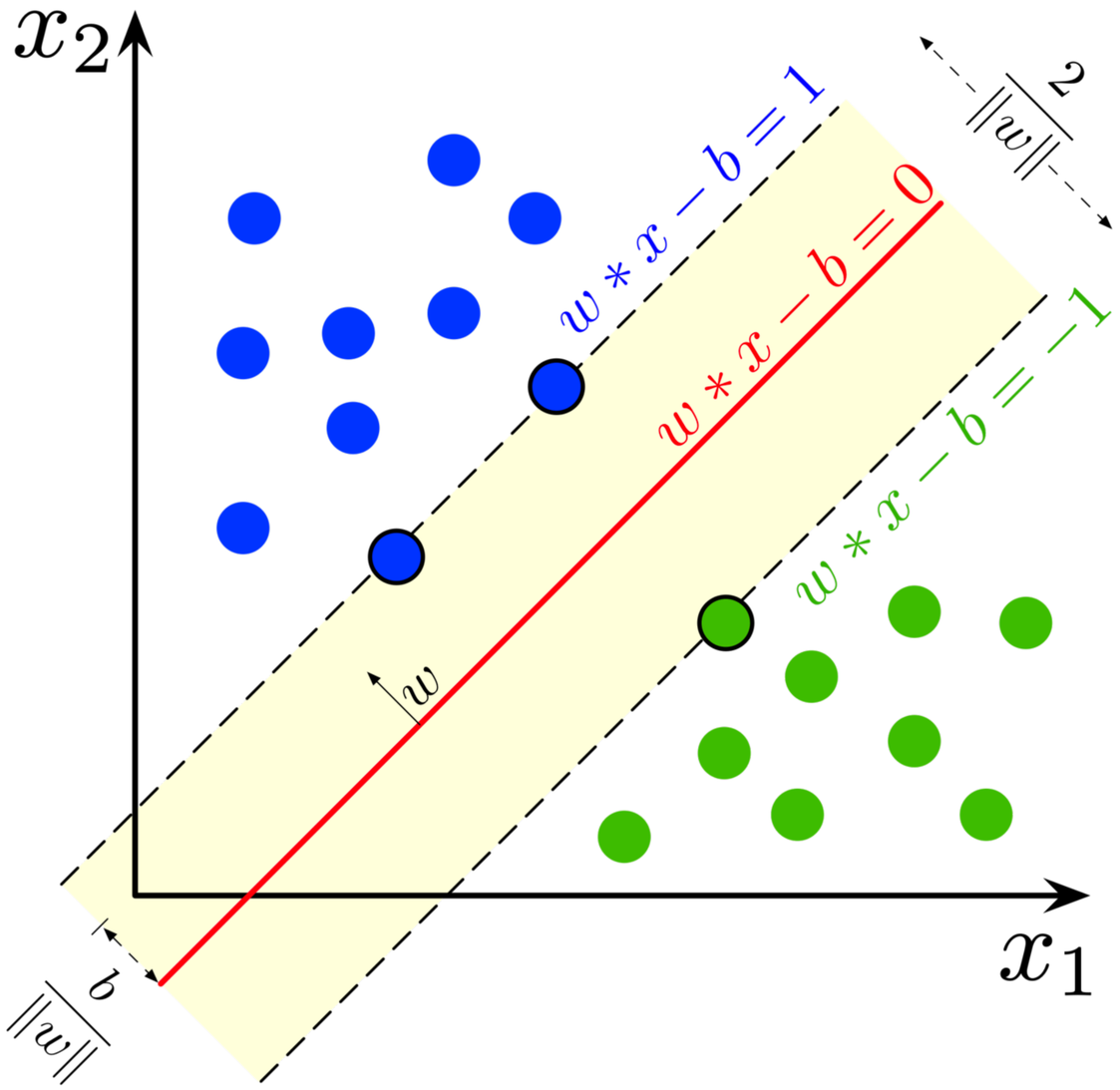
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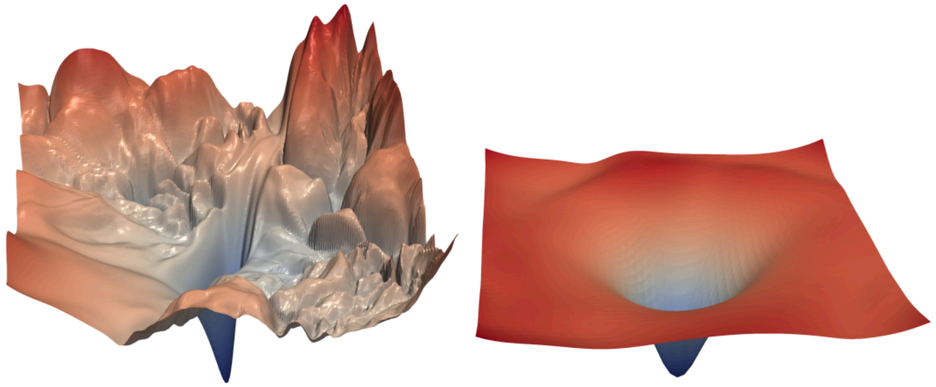
Margin?


x_2
x_1
$w * x - b = 1$
$w * x - b = 0$
$w * x - b = -1$
w
$\frac{2}{\|w\|}$
$\frac{b}{\|w\|}$

Margin of DNNs?

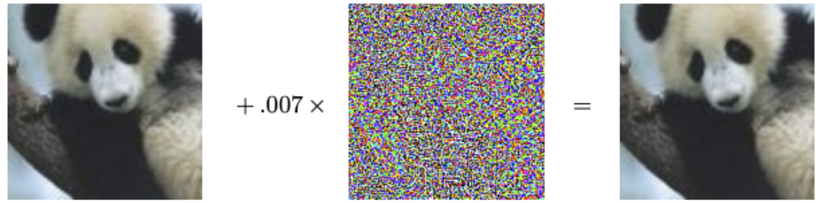
- More advanced topic
- High dimensionality
- Which notion of norm? L1, L2, L-inf?
- At which level?
- Tldr: it’s more complicated

Margin of DNNs?



(a) without skip connections (b) with skip connections

- Also related to loss landscape Flatness
- Which in turn relates to adversarial robustness



$\boldsymbol{x}$
"panda"
57.7% confidence

$\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"nematode"
8.2% confidence

$\boldsymbol{x} + \epsilon\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"gibbon"
99.3 % confidence

Visualizing the Loss Landscape of Neural Nets
EXPLAINING AND HARNESSING ADVERSARIAL EXAMPLES

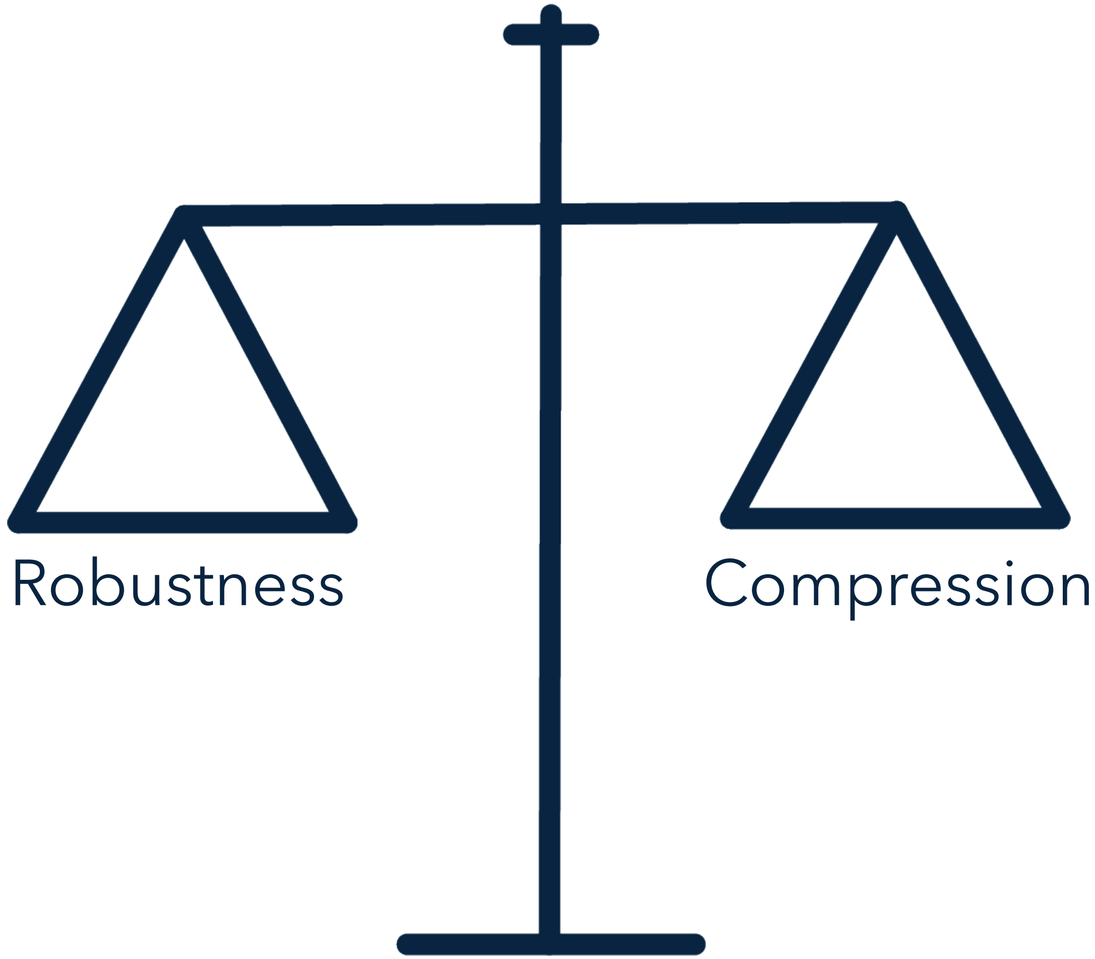
Margin of DNNs?

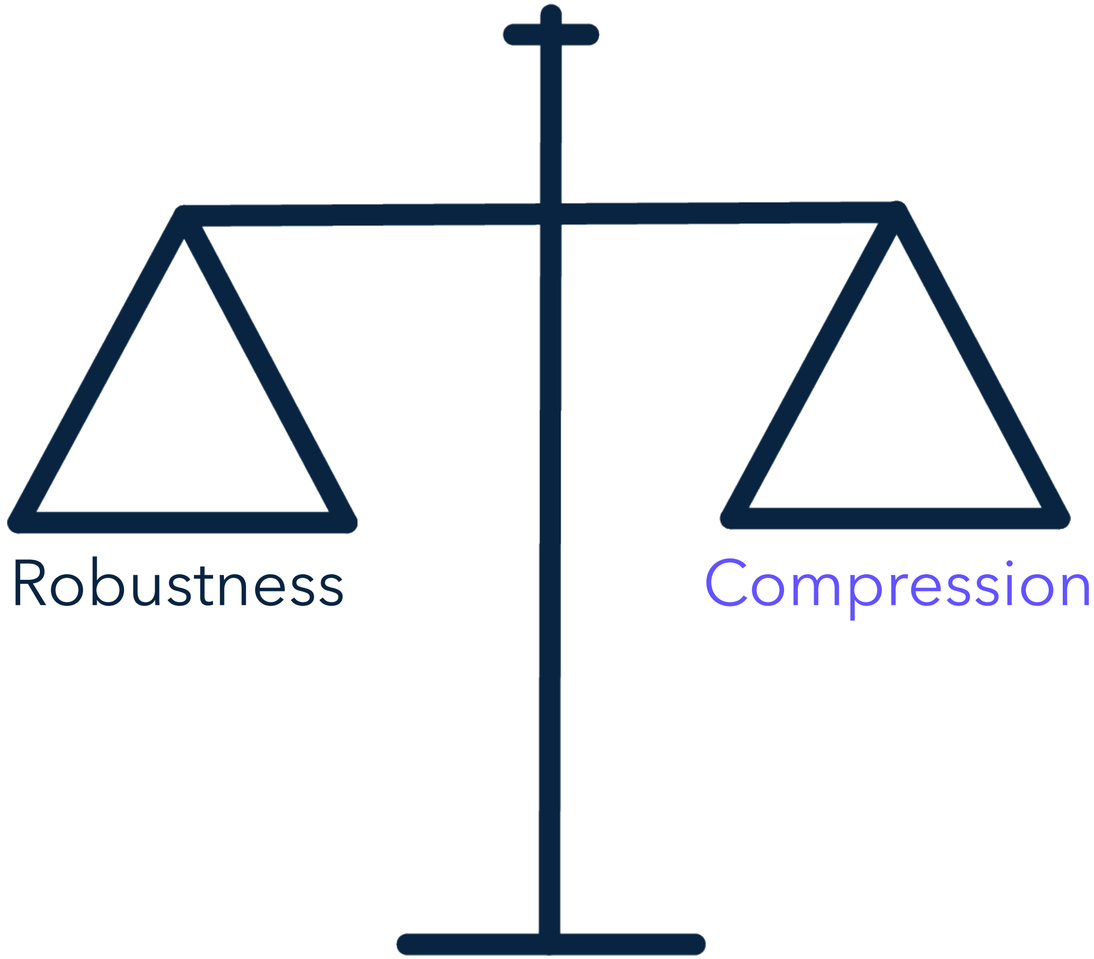
- Also related to loss landscape Flatness
- Which in turn relates to adversarial robustness

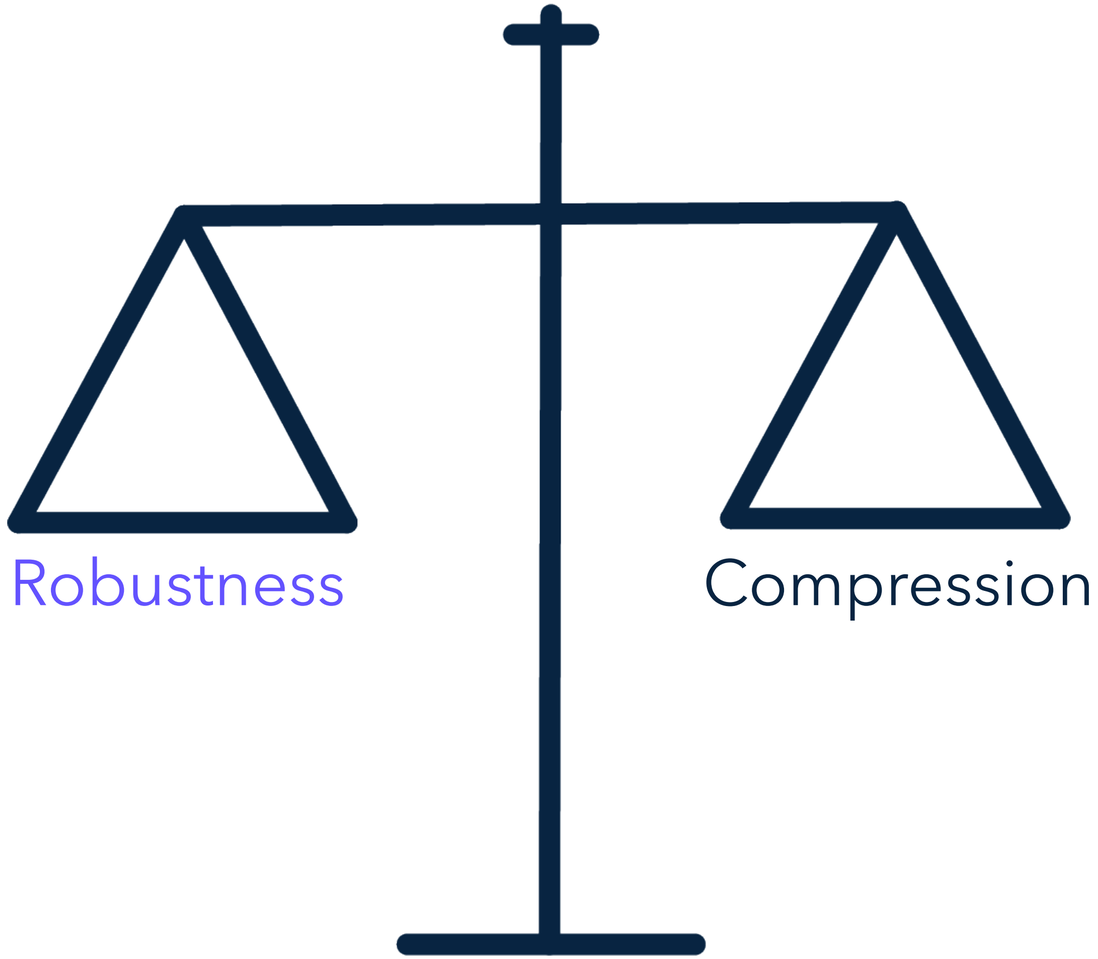
Margin of DNNs?

- Also related to loss landscape Flatness
- Which in turn relates to adversarial robustness
- More importantly: it doesn’t correlate well with generalisation in practice

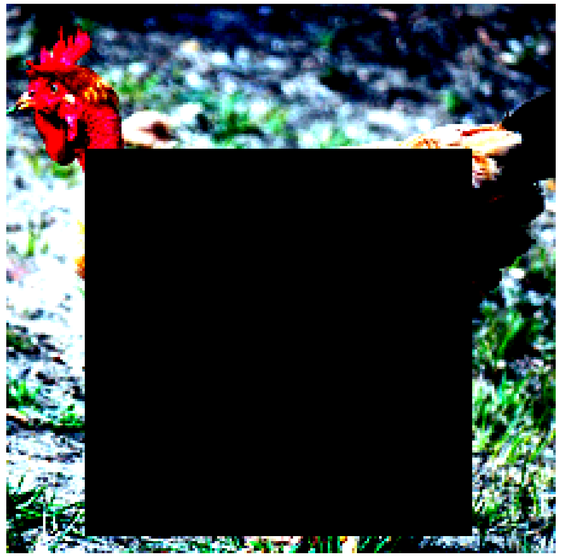
Other notions of hypothesis quality?


Robustness
Compression

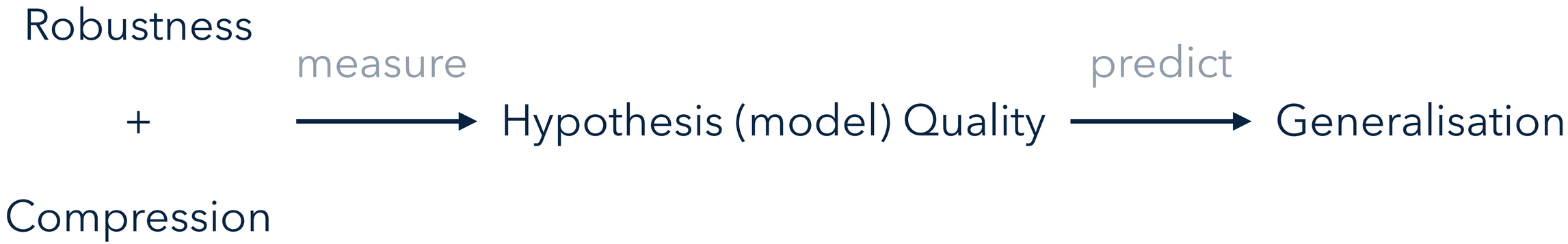

Robustness
Compression


Robustness
Compression

Robustness Example



Putting Things Together



It’s an active area of research

Personal take

Robustness

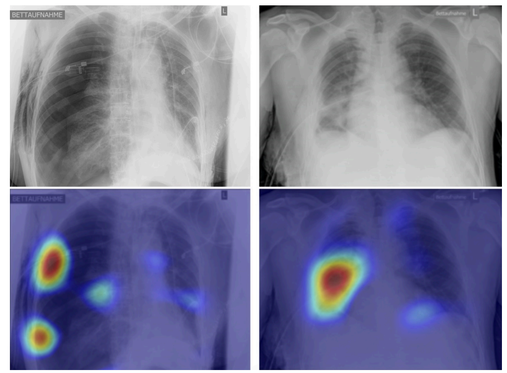
+

Compression

measure → Hypothesis (model) Quality → Generalisation

predict

Model-centric notions of distribution gap



Take-aways

- Model quality is not just accuracy

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- Making machine learning trustworthy is very much work in progress

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Questions?